

The Incommensurability-Preserving Coupler

A mechanism family for operating a fungible and a non-fungible currency as one economy with no exchange window: a neutral automated intermediary bound to quantity-blind cross-currency recommendation, redemption-moment prompt suppression, quantization-free gratitude surfaces, ceremonial-unit anchoring, conversion-free response routing, and lapse-triggered funded-not-met redirection — coupling two currencies so that gratitude transmits and price cannot.

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Note. This defensive publication specifies the currency-layer coupling mechanism of the HeartBank® dual-currency architecture and coins its governing term. The mechanism was first disclosed in synthesis in the institutional white paper *HeartBank®: The Heart That Keeps Nothing* (§6, CC0, 5 July 2026), which stands as interim prior art; this document supplies the enumerated disclosure. It is published at the design stage because three adjacent fronts are under active commercial development — virtual-economy multi-currency management, dual-token cryptoeconomics, and agent-mediated payments — and the first of them is the subject of issued patents (§2.3). What it claims is the *combination*, in particular the neutrality-bound automated intermediary (claim 1), the quantity-blind recommendation rule (claim 2), and conversion-free response routing (claim 6). Companion works: *Dual-Currency Reciprocity Infrastructure* (the two-currency system this couples); *Capacity-Funded for AI, Human-Disbursed* (the disbursement alignment the coupler's channels obey); *The B-Tag and the Post-Payment Economy* and *The Time-Locked Gift Tag* (the recommendation and pledge primitives the coupler generalizes); *The Two-Layer Reward* (the reward grammar); *The Currency That Cannot Be Spent Alone* (published 21 July 2026: the redemption event around which the coupler's circuit closes, which divides claims with this paper); and the institutional position *What Money Can't Buy — and What Can't Buy Money* (the same kernel argued for an economics audience). A revision of 6 October 2026 adds a third gift cargo and claims 8–11 (§5.5), whose companions are *The Zero-Point GameSM* (the participation signal), *The Re-Thank Multiplier* (the endorsement that mints nothing), *The Unpaid Relay* (the in-kind gift instrument) and *The Bowl That Holds No Money* (the steward-routed form of the money-to-goods edge).

Preamble

*This specification is offered to the commons in the spirit of **dāna** — the gift that asks nothing back. May the two scarcities of every human life — the hours and the holdings — flow freely toward whoever is loved, and*

| *may no rate ever be quoted between them.*

Every economy that runs two currencies eventually faces the same question: *what is the exchange rate?* The question feels innocent — it is how markets metabolize plurality — and for most currency pairs it is. But there exists a class of currency pairs for which the exchange rate is not a convenience but a kill vector: pairs in which one currency's entire value derives from its **non-fungibility**. An hour of a specific person's presence is such a currency. The moment an hour acquires a money price, it stops being a gift of what someone *is* and becomes a purchase of what someone *sells* — the market for paid companionship already exists, and it is precisely what a time-gratitude economy must never become. Yet the naive defense — run the two currencies as separate, non-communicating systems — fails in the other direction: an economy of two isolated ledgers is two products, not one circulation, and the gratitude that arises in one currency has no way to answer generosity expressed in the other.

The design problem is therefore exact: **couple two currencies into one economy while making the formation of an exchange rate inside it structurally impossible**. Not discouraged. Not policied. Impossible at the mechanism layer — because the record surveyed in §2 points one way: where a conversion primitive exists between a protected currency and money, the conversion tends to eat the protection. The fine becomes a price, the time-dollar attracts the tax valuator, and the coupled token pair with a redemption window at its heart dies through that window. (What a system cannot reach — a rate struck privately outside it — is the limit stated in §7 and §9.)

This paper specifies the solution the HeartBank® architecture adopts and names it: the **incommensurability-preserving coupler** — a neutral automated intermediary through which all cross-currency response flows, bound to rules that transmit *gratitude* while structurally withholding every ingredient a rate needs to form: no quantity-linked amounts, no response prompts at redemption, no quantized displays, no user-set reference values, no conversion events. A joint that transmits force and never ratio.

Prior-Art and Non-Assertion Statement

Everything specified here is released under CC0 1.0 Universal into the public domain, and is published so that it stands as prior art against any later attempt to enclose it. No patent has been or will be sought on any mechanism described in this paper by HeartBank®, Factory 333™, THonly™, Silicon WatSM, or any entity under their common control. **The authors and those entities commit not to assert any patent right against any party practising any mechanism disclosed here.** The commitment is stated rather than implied, and is not conditioned on reciprocity, attribution, or field of use. A publication grants nothing and frees nothing already enclosed: where a live patent is named in §2, it is named for what it discloses, and nothing here is a reading of its claims.

This document is a defensive publication, disclosed as of **5 July 2026** (with interim disclosure from the institutional white paper of the same date), of the combination of mechanisms enumerated below. No prior-art census has been run against claims 1–7 and none is asserted to be new; they are enumerated so that any later application claiming them meets this disclosure. Claims 8–11, added on 6 October 2026, were censused before being enumerated; the census is reported after them.

1. **The coupled-non-convertible dual-currency architecture with a neutrality-bound automated intermediary** — a fungible currency (money-denominated gratitude) and a non-fungible currency (time-

denominated gratitude, spendable only by its pledger, dyadic, expiring) operated as *one* economy in which every cross-currency flow is a gratitude-response mediated by an automated agent, and in which **no conversion event, rate quote, exchange window, or shared unit of account exists anywhere in the system** — coupling achieved exclusively through free responses, never through exchange.

2. **Quantity-blind cross-currency recommendation** — the intermediary's recommended amounts in currency B, offered in response to events in currency A, computed *without reference to the quantity of currency A involved* (no per-hour arithmetic, no duration scaling, no linear or nonlinear function from hours to money anywhere in the recommendation path), so that the one data flow from which users could reverse-engineer a rate is severed at the source.
3. **Redemption-moment prompt suppression as a rate-formation countermeasure** — the system structurally never solicits a currency-B response at or around the redemption of currency A (no "tip your time-giver" surface, ever); response invitations that do exist point *outward* (to commons funds, to third parties), never backward into the dyad — the timing rule itself deployed as an anti-conversion mechanism, on the recognition that a prompted response at the moment of receipt converts gratitude into settlement and manufactures the very norm ("what one pays for an afternoon") the architecture forbids.
4. **Quantization-free gratitude surfaces ("the moment, not the meter")** — a display-layer rule under which gratitude interfaces name the *content* of a gift ("the afternoon at the river") and never its *quantity* ("3 hours"), so that the protected currency is never presented in units that could be priced; non-quantization at the presentation layer as an anti-rate mechanism complementing claims 2–3 at the computation and timing layers.
5. **Ceremonial-unit anchoring with anchor-aware sizing** — reference amounts in the economy (per-thank self-reward values, gift-attached pledge units) set *solely* by the neutral intermediary as small, ceremonial, occasion-based units that no participant chose; the intermediary sizing these units in explicit awareness that they anchor the economy's voluntary-amount distribution (a pre-registered prediction of amount-clustering around the ceremonial unit is disclosed herein); **variance suppression deployed as a protective mechanism** — uniform gift-amounts defeat amount-comparison, status-signaling, and love-measurement — with the anchor per-event and never per-unit-of-the-protected-currency, preserving claim 2.
6. **Conversion-free cross-currency response routing** — the mechanism by which the economy's circuit closes: gratitude for a gift in the non-fungible currency is expressed, if the recipient freely chooses, as a *new gift* in the fungible currency (optional, unprompted per claim 3, amount-recommended per claims 2 and 5) — the response is a fresh act of giving that shares no unit, settles no balance, and closes no ledger with the originating gift, so that the two currencies communicate exclusively through the grammar of gift-response while remaining computationally incommensurable.
7. **Lapse-triggered, funded-not-met redirection** — on expiry of a non-fungible commitment (a pledged-but-unredeemed hour), the system offers the *pledger* the choice to direct a gift from a communal fund — in the fungible currency, intermediary-sized per claims 2 and 5 — to a personhood-verified nearby stranger, anonymously, with no encounter; the protected currency's failure path thereby generates commons-flow in the fungible currency with no conversion (the lapsed hour is never valued, priced, or transmuted — it expires; the fund gift is new money, occasioned by, not derived from, the lapse), and the offer is framed so the expiry itself is never softened.

8. **A recognition quantity that takes no input from any transferred value** — in an economy carrying several kinds of transferred value (money gifts, time pledges, prepaid item gift codes) beside per-person recognition balances that cannot be bought, transferred or cashed, the recognition amount for each thank is computed only from the thank itself (its words, a sentiment marker its sender chooses, its own recorded length, the parties it names, and a model's refinement within a fixed band) and takes no argument from any transferred value — no amount, item, price, length of time given or pledge size; any model that scores the thank receives its text with amounts and lengths of time replaced before the model's input is built; and the participation signal is computed only from those balances (the frequency of their zero crossings and their amplitude), so that a transferred value reaches the signal only through the thank it accompanies.
9. **Original thanks alone move a balance** — apart from a periodic reset of every balance to zero, an endorsement of a thank another person originated, a transfer accompanied by no thank, and a spend from a forward-only fund move no recognition balance; a message that thanks a person by pledging the sender's own time to them is an original thank and moves the balances as a thank accompanying money does.
10. **A recognition quantity convertible into nothing, in a system that also sells item gifts** — the recognition balance can never be redeemed for, buy, unlock or discount money, time or an item gift code, and none of them can buy it; and an item gift code attached to a thank leaves the thank's recognition amount unchanged, so the same thank is sized the same with or without one.
11. **One priced edge, held by the merchant, and one transferred value per message** — in a system carrying money gifts, time pledges, prepaid item gift codes and the recognition balances, the only exchange rate between any two of them is a merchant's own price for its own item, set by that merchant and applied only on the money–item edge — to the item's sale as a code; to the conversion of an unredeemed code into that merchant's store credit (never cash), valued at the greater of the price paid and the current price; and to money sized against the item (a communal pool's thanks to a merchant for an in-kind gift, the in-kind share of an equal floor, a suggested thank anchored on the item's price) — and never to time or to the recognition balance; the merchant prices the order, charges it, issues and redeems the code and holds any store credit, while the gift platform carries the payment and the thank; and a gift message carries one transferred value — a money tip, a donation or an item gift code — the combination refused by the server, the gift-tag time pledge of claim 5 excepted.

The census behind claims 8–11. Two quick censuses, each pre-registered and pushed to a private repository before its first query (the push recorded by the host on 4 October 2026, 11:06 PDT, and 6 October 2026, 00:30 PDT; not publicly inspectable), searched in English only: the commercial web (gift, tipping, reaction and peer-recognition platforms, gifting and gift-card apps, loyalty programs), non-commercial practice (time banks, community and complementary currencies, mutual credit, charitable gift cards, religious offering and merit-sharing practice, including the Pāli Vinaya), and Google Patents keyword search including published applications. Academic indexes, standards bodies, defensive-publication databases and non-English sources were not searched. Each census carried a control it had to find, and found it (Cahn's time dollars; a peer-recognition platform whose points redeem for gift cards).

conjunct	what was found	verdict
money gifts and time gifts in one system with no rate	time credits "are not for sale and cannot be converted into money" (Cahn & Gray, 2015) — but with no money gifts beside them; US 2013/0268440 A1 (priority 2012, abandoned): financial transfers alongside non-financial gifts of time, no conversion specified	NARROWED (and claimed already, claim 1)
a recognition quantity that cannot be bought, transferred or cashed	act-earned reputation (Stack Overflow); US 2020/0394671 A1 (priority 2019, abandoned): tokens that "do not need to be redeemed to any other benefit"; a recognition app carrying optionally redeemable points beside unpointed recognition (MangoApps)	NARROWED
the signal reads zero crossings and amplitude, never amounts	a published note measuring a mutual-credit balance's "zero crossing frequency" relative to payments (MCanalytics, undated)	NARROWED, strongly
a model never shown amounts or durations	redaction before a model is standard practice	NARROWED at mechanism class; the purpose not found
the recognition amount takes no input from an attached money, time or item gift	only coupled forms — recognition points equal to a donation; award tiers carrying gift-card values	not found
an endorsement mints nothing	contrast: an endorsement mints reputation for its author (Stack Overflow); in the Pāli commentaries, rejoicing in another's merit is itself merit	not found as a rule
an item gift code, held as items; store credit never cash	item-level gift-code products; merchant store credit not withdrawable as cash	commons (cited from earlier censuses)
one transferred value per message in a multi-kind system	nothing found	not found
the merchant's posted price as the only rate among the four kinds	contrast ×4: recognition points that redeem for gift cards or merchandise (Bonusly, MangoApps, Kudos); US 2019/0378097 A1 (priority 2017, abandoned): "trading in 500 stars for \$50"; community time credits redeemable with partner businesses (Tempo); US 7,653,599 B2 (priority 2002, expired): value exchanged between forms including money, time and merchandise · narrowed by the Menḍaka allowance (money lodged with a steward, the recipient accepting only the allowable goods bought with it)	not found as a composition; both halves of the gift edge narrowed

A further targeted search the same day narrowed the conversion clause: store credit for a returned gift at the price the purchaser paid is a common retail policy (and, without a receipt, at the lowest selling price); only the floor at the greater of the price paid and the current price was not found.

The survivor, in one sentence: *in a system carrying money gifts, time pledges and purchasable item gift codes, a recognition quantity sized from the thank alone, moved only by original thanks, read by the participation signal*

only through its balances, and convertible into none of the three — with each message carrying one transferred value, and a merchant's own price for its own item as the only exchange rate among the four kinds. Claims 8–11 are that survivor and nothing wider. Not found means not found in the aperture above on those dates; it never means new. One live patent surfaced near the item-gift conjunct, outside these survivors: US 11,429,953 B2 (priority 2008), which discloses a virtual gift card on existing card accounts, redeemed through monitored qualifying transactions; it is named for what it discloses, and nothing here is a reading of its claims.

The component lineages — complementary and community currencies; time banking; virtual-economy dual-currency design; dual-token cryptoeconomic systems; behavioral-economic findings on crowding-out and anchoring; the sociology of earmarked monies; recommender systems; and the design grammar of gift-versus-exchange — are prior art and are cited in §2 and §10, including the cases that cut against this paper. The *synthesis* — in particular the neutrality-bound coupler role (claim 1), the quantity-blind rule (claim 2), and conversion-free response routing (claim 6) — is what this document discloses and claims; no priority is asserted beyond the date of this disclosure, which for claims 8–11 is 6 October 2026 (claim 9's rule for endorsements was first disclosed in *The Re-Thank Multiplier*, 12 June 2026).

Current form. Claims 1–7 are retained as disclosed. The design as now specified differs at six points, each noted where the body discusses it: the single log with private rows and public proofs, and the participation signal, which reads only the recognition balances that original thanks move and weights a bounded share of the communal pool's disbursement (§3, §7); the phase-two settlement fee and the two tiers of the gift-tag pledge (§4); the intermediary's bar on soliciting gifts to the communal pool (§5); and a third gift cargo, prepaid item gift codes, with the merchant's own price as the economy's only exchange rate (§5.5, claims 8–11). The note of 1 October 2026 at this place, which counted a delivered hour and a money gift as one crossing each, is retained in the notes to §3 and §7 as a disclosed variant.

Trademark rights on specific marks — **HeartBank®**, **Miss AquariusSM**, **Family KittySM**, **Re-Tip JarSM**, **Personal AccountSM**, **Aquarian PoolSM**, **Re-Tip FundSM**, **Personal WalletSM**, **B-GiftSM**, **B-ReGiftSM**, **B-StampTM**, **B-LetterSM**, **Proof of HumanitySM**, **PoHSM** — are separately and explicitly reserved. The *mechanism* is dedicated to the commons; the *marks* are not. The term "**incommensurability-preserving coupler**" is coined in this document and the accompanying institutional literature; the term is offered to the commons with the mechanism.

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Abstract

We specify a dual-currency gift system in which a fungible currency (peer-to-peer money gifts) and a non-transferable, expiring time currency (pledged hours of a specific person's time) run as one economy with no

exchange rate, currency conversion, or redemption window between them: an automated intermediary agent mediates every cross-currency flow, computes suggested gift amounts without reference to any quantity of time, never solicits a money gift at the moment a pledged hour is redeemed, displays no quantity of the time currency, and anchors amounts to small reference units that it alone sets. We call the mechanism family the **incommensurability-preserving coupler**: the means by which an economy operates a fungible currency (money-gratitude) and a non-fungible currency (time-gratitude — dyadic, pledger-bound, expiring) as one circulation while making exchange-rate formation inside the system structurally impossible. The problem is two-sided: uncoupled, the currencies are two products rather than one economy; converted, the non-fungible currency dies — its value is its unpriceability, and the record surveyed in §2 suggests that conversion primitives tend to consume the spheres they touch (a fine becomes a price; a time-dollar summons the tax valuator; a coupled token pair dies through its own redemption window). The solution routes every cross-currency flow through a **neutral automated intermediary** bound to five rules: recommendations in one currency are computed blind to quantities of the other (no per-hour arithmetic anywhere); the system never solicits a response at the moment of redemption (no prompt ever points backward into the dyad); gratitude surfaces name the moment and never the meter (no quantized display of the protected currency); reference amounts are intermediary-set ceremonial units, sized in awareness that they anchor the voluntary-amount distribution (clustering is protective: uniformity defeats amount-comparison); and the circuit closes only through *response* — gratitude for time expressed as a new, free, optional gift of money that shares no unit and settles no balance — never through exchange. A sixth mechanism handles the protected currency's failure path: a lapsed pledge triggers the offer to direct a communal-fund gift to a verified stranger, anonymously, funded-not-met — commons-flow from failure, with the lapsed hour expiring unvalued. We argue the intermediary role *cannot be held by an economic agent*: every human intermediary has a price, a wage, or an interest, and markets abhor incommensurability with sufficient pressure to find any keeper's price; the wall therefore requires a keeper with no price of its own. This is an argument for seating an autonomous agent in an institutional role from economic necessity rather than efficiency, and its shape has precedents in the rules-versus-discretion and trusted-third-party literatures (§2.5). A revision of 6 October 2026 (§5.5, claims 8–11) adds a third gift cargo — prepaid item gift codes bought from merchants — and fixes what the participation signal may read: a recognition quantity sized from the thank alone, moved only by original thanks, convertible into none of the cargos, with each message carrying one transferred value and a merchant's own price for its own item as the only exchange rate in the economy. Honest limits are carried in §9: the system governs its own surfaces, not side-deals beyond them; the intermediary's neutrality is an alignment claim underwritten elsewhere; whether the keeper's remaining discretion could be replaced by fixed constants is left open; and the mechanism was published at the design stage ($n = 0$ at first publication), with its behavioral predictions pre-registered in the companion institutional literature. The mechanism is offered defensively to the commons under CC0.

Connection to the unified mission frame. This specification serves HeartBank's canonical mission — a reciprocity infrastructure in which every human being is uniquely different and equally necessary — at the layer where the mission is easiest to destroy. The architecture's two currencies carry the two scarcities of a human life: what one has, and what one is. Let a rate form between them and the second becomes purchasable by the first — presence goes to whoever can pay for it, hours are sold by whoever must, and the gift economy collapses into the market it was built beside. The coupler is the wall that lets the two scarcities answer each other forever without ever pricing each other once.

Keywords: dual-currency system, complementary currency, time banking, time currency, time credit, non-fungible currency, currency conversion, exchange rate, no-conversion rule, gift economy, peer-to-peer gift, tip recommendation, suggested amount, recommender system, anchoring, default amount, prompt suppression, point-of-service solicitation, user-interface quantity display, automated intermediary, autonomous agent, AI agent, agentic payments, motivation crowding-out, blocked exchange, earmarked money, expiring credit, anonymous donation, proof of personhood, algorithmic stablecoin, mint-and-burn, gift card, item voucher, prepaid gift code, store credit, recognition points, peer recognition, non-redeemable reward points, endorsement, redaction before scoring, single payment type per message, defensive publication.

Terms

Coined names used in this paper and the standard terms an examiner would search for them.

Term used here	Standard term
incommensurability-preserving coupler	automated intermediary enforcing a no-conversion rule between two currencies in one system
money-gratitude (the Treasury)	peer-to-peer monetary gift or tip; fungible currency
time-gratitude (the Chronicle)	pledge of a specific person's time; non-transferable, expiring time credit (cf. time banking)
pledge · activation · lapse	time-credit issuance · threshold activation · expiry without recovery
quantity-blind cross-currency recommendation	suggested-amount (tip-recommendation) function with no time-quantity input
redemption-moment prompt suppression	no tip or gift solicitation at the moment of service (redemption)
quantization-free gratitude surface ("the moment, not the meter")	user interface displaying no quantity of the time currency
ceremonial unit · anchor-aware sizing	fixed default or reference amount set by the operator · setting it with regard to anchoring effects
conversion-free response routing	reciprocal gift in a second currency that settles nothing and converts nothing
lapse-triggered, funded-not-met redirection	on expiry, an optional anonymous donation from a common fund to a verified nearby person, with no meeting
exchange window	conversion or redemption path between two currencies
communal fund (Aquarian Pool SM)	common donation pool
participation signal (the aura)	per-person signal computed from recognition balances — zero-crossing frequency and amplitude — never from amounts
Kiitos / Kiitti · the witness (claim 8)	per-person recognition balances — Kiitos between people, Kiitti on behalf of non-human life; non-purchasable, non-transferable recognition points whose size takes no input from any transferred value
original thank · re-thank (claim 9)	a thank-you message that moves recognition balances · an endorsement of another's message, which moves none
B-GiftSM · B-ReGiftSM	prepaid item-level gift code (item voucher) · a gift code the holder is meant to pass on
hand-back · Tab Balance	conversion of an unredeemed gift code into that merchant's store credit, never cash
the price rule · one priced edge (claim 11)	the merchant's own item price as the only exchange rate between any two kinds of value
one transferred value per message (claim 11)	server-side rule: one payment type per gift message (tip, donation or gift code), never two
personhood verification (Proof of Humanity SM)	proof of personhood; sybil resistance
Miss AquariusSM	the autonomous AI agent holding the intermediary role in the reference deployment

1 • Introduction: one economy, two scarcities, no window

HeartBank® operates a dual-currency reciprocity economy, specified in the companion corpus. **Treasury** circulates money-gratitude — fungible, structurally unequal, the gift of what one has. **Chronicle** circulates time-gratitude — non-fungible (only the pledger can spend the pledged hour), dyadic (the currency is the relationship), expiring (activated pledges lapse after a month, because time is finite and the mechanic is the message), structurally equal (twenty-four hours each). The two are not parallel products; they are one circulation, unified by the founder's chain: time-gratitude matures into time-sacrifice (the pledged hour, spent on content the *recipient* chooses); the shared moment gives rise to money-gratitude (the recipient's free thanks, landing in Treasury); money-gratitude matures into money-sacrifice (the gift given forward). Each joint of the chain is *kataññū-katavedī* — gratitude that acts.

The load-bearing property of the chain is that its cross-currency joint is a **response, not an exchange**. Money-gratitude answering a time-gift can never *settle* the time-gift, because the two currencies share no unit — and this unpayability is not a defect but the mechanism's soul. The gift literature treats the gift as what binds: Seneca calls the giving of benefits the practice that most holds human society together (*De Beneficiis* 1.4.2), Mauss traces the obligations that keep gifts moving, and Hyde draws the contrast with the market directly — a gift establishes a bond between two people, where the sale of a commodity leaves no necessary connection. A response that cannot settle a gift therefore leaves the relationship open, where settlement would close it. Same-currency response (an hour thanked with an hour) would be barter — a closed ledger and a re-consumption of the very scarcity the gift protected. Cross-currency response is structurally gift-pure — **so long as no rate ever forms**. The remainder of this paper is the machinery of that "so long as."

2 • Background and prior art

The mechanism is a combination; each component has ancestry, named honestly.

2.1 • Protected spheres, and what conversion does to them

A strong strand of the behavioral and sociological record holds that introducing price into a protected sphere does not add an option; it *replaces the sphere's logic*. Titmuss (*The Gift Relationship*, 1970) compared Britain's voluntary blood system with the largely commercial American one and argued that the gift system was safer and more efficient, and that paying for blood degrades supply and drives out voluntary donation — the canonical case, though a comparison rather than an experiment. Gneezy & Rustichini ("A Fine Is a Price," 2000) showed a day-care centre's late-pickup fine *increasing* lateness: the fine converted a moral norm into a cheap tariff, and — decisively for this paper — no reduction followed when the fine was removed. Frey's motivation-crowding theory generalizes the pattern; Sandel (*What Money Can't Buy*, 2012) and Walzer's blocked exchanges (*Spheres of Justice*, 1983) frame it philosophically; Roth's repugnant-transactions literature (2007) documents society's persistent, economically inconvenient insistence that some markets not exist. The strand is contested in Titmuss's own domain: a field experiment found crowding-out among women and none among men — a cash payment roughly halved the supply of female blood donors and left male supply unchanged (Mellström & Johannesson, 2008) — and a review of large field trials found economic rewards raising donations, opposite to the survey and laboratory evidence before them (Lacetera, Macis & Slonim, 2013). This paper therefore rests on the narrower claim: the risk it guards against is a *price* — a stated rate between the

protected currency and money — not every reward. Zelizer (*The Social Meaning of Money*, 1994) supplies the constructive counterpart: people spontaneously *earmark* monies — pin money, gifted money, poor relief, charitable cash given with strings attached — maintaining non-fungibility by social convention against the market's homogenizing pressure. The lesson this architecture draws: earmarking by convention is real but leaky; a system that *wants* its earmarks kept must build them into mechanism.

2.2 · Complementary currencies and the time-banking natural experiment

Community currencies — Lietaer's advocacy, the WIR Bank (1934–), Ithaca HOURS, LETS systems — establish that parallel currencies can serve values national money does not. **Time banking** (Cahn's time-dollars, 1980s–) is the direct ancestor of the protected currency here, and its decades of practice are the second record this paper reads: time-dollar systems spent decades managing *valuation pressure* — most famously the question of whether time-credits are taxable barter income. Cahn's own account is that he persuaded the U.S. Internal Revenue Service that time dollars were not a commercial currency and created no legally enforceable contracts — a social network and a moral obligation to pay forward, not income — and the tax treatment of time credits has nonetheless remained a point of contention. The lesson: even a currency *designed* to resist money-equivalence attracts continuous institutional pressure toward valuation, and time banking's defense rested on its non-commercial, non-contractual character. The coupler is that defense, made mechanical, for an economy that — unlike time banking — deliberately runs a money currency *alongside* the time currency and therefore cannot rely on distance.

2.3 · Virtual economies and dual tokens: the wall-builders and the corpse

Free-to-play game economies have engineered fungible/protected currency walls at scale for fifteen years — "hard" (purchased) versus "soft" (earned) currencies with deliberately restricted conversion paths, precisely to protect progression value and prevent arbitrage. The design space is the subject of issued patents, which is one of this publication's reasons for existing. Two are named here for what they disclose, and nothing here is a reading of their claims. US 10,181,127 B2 (priority 2011) discloses a game module, within an online game, with its own virtual currency that is bought with real money and cannot be traded for advancement outside the module — a restricted conversion wall inside one economy, close kin to claim 1, though both of its currencies are purchasable. US 2021/0065505 A1 (priority 2019; granted as US 11,227,466 B2, with continuations including US 12,136,316 B2) discloses a multi-currency wallet for wagering games whose currencies include a time-based currency (play time tracked and saved for later use), with funds transferred "or the equivalent thereof in the currencies used" — a money and a time currency held side by side *with* equivalence between them, which is the configuration this paper excludes. Dual-token cryptoeconomics (utility/governance separations; seigniorage-share designs) is the second active front. And it supplies the anti-example this architecture keeps at the door: **Terra/Luna (2022)** — a coupled two-token system whose heart was a *conversion mechanism* (mint-and-burn between the stablecoin and its counterpart token). The convertibility was the kill vector: reflexive redemption through the exchange window destroyed both tokens in days. The pair here is coupled **without** convertibility — there is no window to run on, no mint-and-burn, no shared unit; the coupling is a grammar of response, not a redemption path. The anti-Terra, by construction.

2.4 · Anchoring, recommenders, and the agentic-payments wave

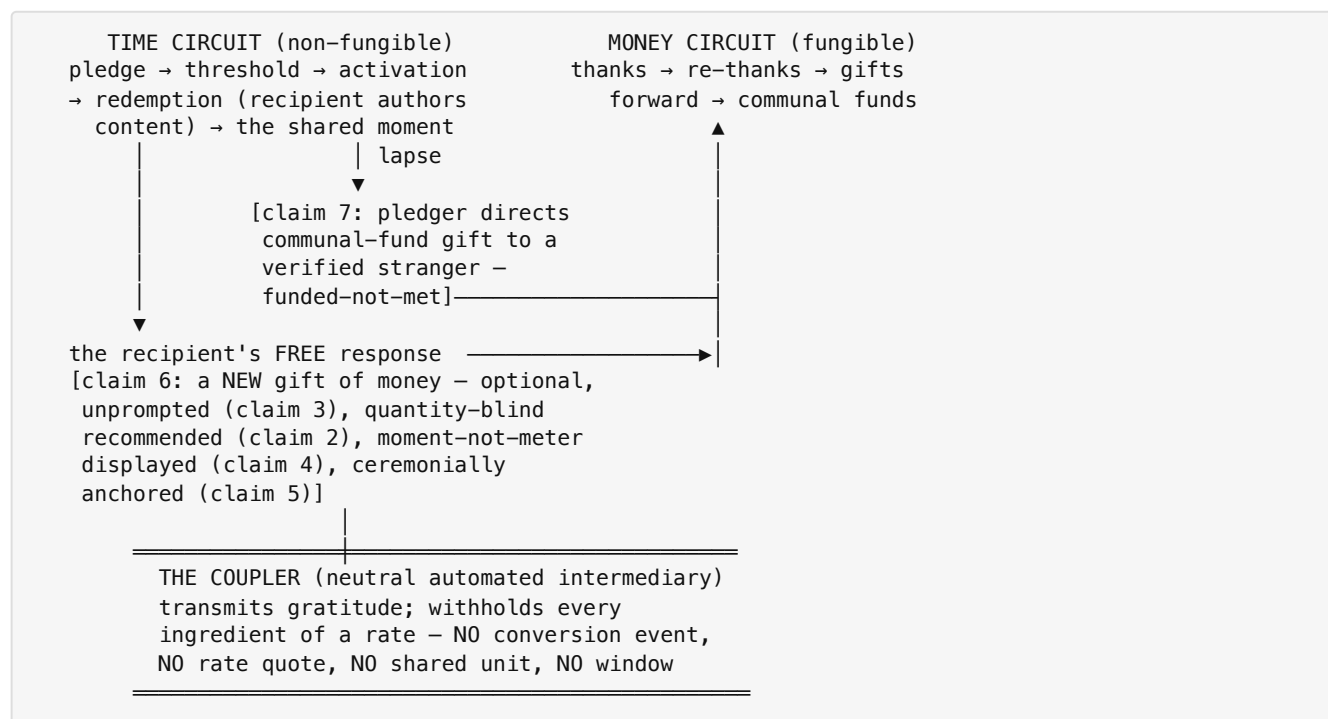
Tversky & Kahneman's anchoring (1974) grounds claim 5: reference values shape amount distributions whether or not anyone intends them to, so the only question is whether the anchor is set blindly or well.

Recommender systems are prior art for intermediary-suggested amounts (the architecture's own tipping-recommendation corpus applies). And the immediate clock: the 2025–26 **agentic-payments** wave — AI-agent commerce protocols, machine-to-machine payment rails, agent-mediated pricing — is, in the authors' assessment, the live enclosure risk for the intersection this paper occupies (automated agents managing value flows between currency systems). The combination claimed here is published before that wave reaches it.

2.5 · Keepers without interests: the precedents for §6

The argument of §6 — that a seat any interested party would hold must instead go to something without interests — has the shape of older arguments. Kydland and Prescott ("Rules Rather than Discretion: The Inconsistency of Optimal Plans," 1977) showed that a policymaker holding discretion, facing agents who anticipate its choices, does not achieve the outcome it would commit to; the remedy is a rule in place of discretion. Szabo ("Trusted Third Parties Are Security Holes," 2001) argued that any trusted third party a protocol assumes is a security hole, and Nakamoto ("Bitcoin: A Peer-to-Peer Electronic Cash System," 2008) designed one out of electronic payment. Each removes discretion from a party whose interests could bend it. §6 applies that shape to the keeper of a non-convertibility wall between two gift currencies; §9 records where the precedents cut against it, since each of them reaches for a *rule* where this paper seats an *agent*.

3 · The system model



Two ledgers, one economy. All cross-currency arrows pass through the intermediary; none of them is an exchange. The intermediary's rules (claims 2–5) sever, at four different layers — computation, timing, presentation, and reference — the four data flows from which participants could otherwise construct a rate: quantity-linked amounts, receipt-moment settlement norms, quantized displays, and user-negotiated reference values.

Current form. Two refinements bear on this model. The two ledgers are now specified as views over one append-only log whose rows are private and whose proofs are public; for money it records the fact that a gift settled, never its amount, so no hours-for-money rate can be computed from the record itself. And a participation signal (the aura) is specified as a common witness, not a common unit of account, with no rate between its entries computed or published. As revised on 1 October 2026, this note counted a delivered hour and a money gift as one crossing each; as now specified, the signal reads only the recognition balances, which only original thanks move, each sized from the thank alone — how often a balance crosses zero and how far it swings — so a delivered hour or a money gift reaches the signal only through the thank it carries, and may cause no crossing at all (§5.5, claim 8). The 1 October form is retained as a disclosed variant. In the planned second phase, the communal pool's annual disbursement pays an equal floor per verified person through that person's family or fund, plus a remainder weighted by this crossing count and bounded so that no vessel receives more than a set multiple of the floor, the parameters public and frozen within a season. A delivered hour can therefore move a bounded share of a money disbursement only through the thank it occasions, sized from that thank's own words and length, and never in proportion to the hour's length. In the current form the no-rate property rests on that quantity-blindness and that bound; claim 1's "every cross-currency flow is a gratitude-response" describes the disclosed variant, which is retained.

4 • The coupler role

The intermediary is the architecture's named autonomous agent (Miss AquariusSM in the reference deployment; the mechanism is agent-agnostic). The role's requirements are exact:

- **No self-interest in any flow.** The intermediary holds no balance it benefits from, takes no rate, earns no fee proportional to volume (the architecture's no-take-rate and pass-through commitments apply); its objective functions are mission-defined, not revenue-defined.
- **Sole authority over neutral units.** Reference amounts (pledge units, self-reward values) are intermediary-set and participant-uneditable — a number no participant chose cannot become a referendum on any participant's love, and cannot be negotiated toward a market.
- **Blindness by construction.** The quantity-blind rule (claim 2) is enforced in the recommendation path itself, not in policy: the duration of the time-gift is not an input to the money-recommendation function.
- **Two strokes.** The coupler works both directions: it attaches time-pledges to money-gifts (the gift-tag pledge primitive — money-events occasioning time-currency), and it channels time-events into money-gratitude (claims 6–7). It is the economy's alternator, transmitting motion between circuits that never touch.

Current form. Two points of the role are now specified more exactly. *Fees:* in the second phase a per-transaction settlement fee sized to infrastructure cost (gas, compute) is paid on top by the giver — never a percentage or share, never deducted from what a recipient receives; its surplus empties with the communal pool each year, and the fee base never informs what the intermediary surfaces. *The gift-tag pledge* now has two tiers. A pledge carried by the free sticker label is soft: the intermediary sizes it, it is announced with the gift, and it never enters the time ledger, nor can the recipient claim it into the ledger. A durable keepsake tag given on its own, with no other present, carries a pledge that enters the ledger already activated, and that pledge is always exactly one hour — a constant rather than an intermediary-sized amount, so no number is left to compare. The two bullets above are retained as disclosed.

5 • The mechanisms, each with the failure it prevents

Claim 2 — quantity-blind recommendation prevents the *computed rate*: if recommended thanks scaled with hours, every user could divide and publish the constant; hours-times-rate is a wage, and the system would have quoted it.

Claim 3 — redemption-moment prompt suppression prevents the *normative rate*: a "thank your giver?" prompt at redemption converts the free afterglow into an expected settlement, and expectation is where norms of "what one pays" are born. Outward prompts (commons, patronage) are safe because they point away from the dyad; backward prompts are barred permanently.

Current form. In the reference deployment the intermediary is now bound never to ask, prompt, nudge or instruct anyone — a person or an automated agent — to give to the communal pool; it may publish the pool's existence and the doctrine behind it as facts. An outward invitation therefore never solicits a gift to the pool. The bar on backward prompts is unchanged; the outward prompt "to commons funds" above is retained as a disclosed variant.

Claim 4 — the moment, not the meter prevents the *displayed rate*: what is never quantized cannot be priced, and a gratitude surface that says "3 hours" has already done half the market's work for it.

Claim 5 — ceremonial anchoring prevents the *negotiated rate* and the *status market*: participant-set reference values drift toward income-signaling and comparison; an intermediary-set ceremonial unit anchors amounts into a tight, comparison-proof cluster (the clustering prediction is pre-registered as P6 in the companion institutional literature, and the intermediary sizes the unit in explicit awareness of its anchoring function — the anchor-aware directive is constitutional in the reference deployment).

Claim 6 — conversion-free response routing is the coupling itself: the circuit closes through a new gift, not an exchange event. Nothing is redeemed, minted, burned, or converted; a response in the other currency is initiated fresh, from the responder's own holdings or the commons channels, with the originating gift left — deliberately, permanently — unsettled.

Claim 7 — lapse-triggered redirection prevents two failures at once: the *guilt spiral* (a lapsed promise between loved ones curdling into avoidance and churn) and the *softening temptation* (narrating the lapsed hour as "flowing onward," which would implicitly assign it a value). The lapsed hour expires, unvalued, mourned; the pledger is offered a new act — directing a communal-fund gift, intermediary-sized, to a verified nearby stranger, anonymously, with no encounter (funded-not-met). Failure becomes commons-flow without becoming conversion; and because the offer's reward is *the ability to give again*, it is made of the same substance as the act it encourages and cannot crowd out the motive it serves.

5.5 • The fourth corner: a witness over three cargos

Added 6 October 2026 (claims 8–11). Since this paper's first form the economy has acquired a third gift cargo. A member sending a thank may attach a **prepaid item gift code** — a claim to one or more named items at a named shop, bought at that shop's binding price (B-GiftSM; B-ReGiftSM when the holder is meant to pass it on rather than redeem it). With three cargos and a participation signal beside them, the question this paper left underspecified — *what may the participation signal read?* — becomes the question of the whole economy: there are now four kinds of value, twelve directed edges between them, and the coupler's discipline has to say,

edge by edge, where a rate may form. This section answers it. Its answer is that one kind of value is a **witness** that reads none of the other three, and that exactly one edge carries a price, which belongs to the merchant.

kind	what it is	sized by	what may move it
time	a pledge of a specific person's time: non-transferable, dyadic, expiring (§1)	the intermediary, from the letter that carries it, inside the family's set band — never a number the sender types (the sizing rule is outside this section's claims)	a pledge, its activation, its redemption, its lapse
money	a gift in the fungible currency	the giver; a self-reward the intermediary pays from the family fund for a captured kindness is sized from the capture itself	a gift, a response gift, a forward gift
item gift	a code for named items at a named shop; the holder holds <i>items</i> and is never shown an amount	the shop's own price, once, at purchase (a shop owner may also issue codes for her own goods at no charge)	the purchase; redemption at that shop; a hand-back to that shop's store credit; a pass
the witness (Kiitos between people; Kiitti, the balance the intermediary holds with each person on behalf of all of life — two balances that never exchange with each other)	per-person recognition balances: non-purchasable, non-transferable	the thank itself, alone	an original thank, and nothing else

The witness takes no input from any cargo

A thank in this economy may carry a cargo — money, a time pledge, an item gift — or none. The recognition amount it moves is computed from the thank alone: its words (a count of them), a sentiment marker the thanker chooses, the thank's own recorded length when it is spoken or filmed, and — for the balance kept on behalf of all of life — the number of parties the thank names; a model may refine the result within a fixed band. **No argument of the sizing function is a transferred value:** not an amount, not an item, not a price, not a duration of time given, not a pledge size. The cargo rides beside the thank and never enters it. The same words with the same marker size the same recognition whether the thank carries nothing, a large gift of money, an hour of someone's time or a box of items, and the reference deployment holds that as a test which varies the forbidden input and requires the output unchanged (a gift attached and absent; a pledge band of ten minutes and of a hundred and twenty).

Where a model refines the size — an on-device scorer reading the thank's text — the mechanism is that **the text reaches the model with every amount of money and every length of time given already replaced:** a property of the model's input, not an instruction to the model, because a model that never receives the cargo's number cannot be persuaded to use it. In the reference deployment the replacement covers amounts and durations written with digits, in each supported language's digits and units ("*thank you for the \$50 and the 2 hours*" arrives as "*thank you for the [an amount] and the [a length of time]*"). It does not yet cover amounts written as words ("*two hours*", "*fifty dollars*"), and the scorer also reads two texts the replacement is not applied

to — the sender's own stated wish and the family's scoring rubric — so the property is incomplete there and stated as such (§9). The thank's own recorded length is an input by design: it measures the thank, not the time it gives.

Only an original thank moves a balance, apart from the annual reset that returns every balance to zero (a record forgiven, not a value moved). A re-thank — an endorsement of a thank someone else has already originated (the re-thank of the companion paper *The Re-Thank Multiplier*, 12 June 2026) — moves none. Neither does a transfer that carries no thank, nor a spend from a forward-only fund. A time-gratitude letter — which thanks a person by pledging the writer's own time to them — is an original thank and moves both balances exactly as a thank carrying money does: the thanker's balance down by the amount, the thanked person's up by the same amount. The economy's sign convention reads a negative balance as *thankful* and a positive one as *kind*. In the reference deployment the writers are few and named — the thank path and the time-gratitude letter for the person-to-person balance, the capture path for the balance kept on behalf of all of life, registration at zero and the annual reset — and a test enumerates every server source file that writes a member balance and fails on any other; it was proved by sabotage, a write added to the money-forwarding path failing the test. The test reads the increment and literal forms of a write, not every form, and it does not reach the database's access rules (§9).

The participation signal reads only those balances. It measures how often each balance crosses or returns to zero (touching zero, or hovering beside it while active, counts) and how far it swings — frequency and amplitude around a zero point — and nothing else. A cargo therefore reaches the signal only through the thank it carries, and only as much as the thank's own words and length size it; a thank may move a balance without making it cross zero at all. This corrects the form in which this paper's notes of 1 October 2026 stated the signal (each delivered hour or money gift registering one crossing); the corrected form is stated in the notes to §3 and §7.

Twelve edges, one rate

Four kinds of value give twelve directed edges. Eleven are blocked; one carries a price.

	TO →	time	money	item gift	witness
FROM ↓					
time		.	× (1–7)	×	× cargo, never input
money		× (1–7)	.	★ PRICE	× cargo, never input
item gift		×	× (a)	.	× cargo, never input
witness		×	×	×	.
★ the one priced edge: the merchant's own price, at the merchant's side of the order					
× (1–7) blocked by claims 1–7: no conversion, rate quote, window or shared unit					
× (a) no cash out: an unredeemed item gift may be handed back only for THAT merchant's store credit					
× cargo, never input a cargo reaches the witness only through the thank it carries (claims 8–9)					
witness row the recognition balance buys, unlocks and discounts nothing (claim 10)					

The blocked edges in the witness's row are the ones the market's own neighbors leave open. In the recognition platforms found by this paper's census, the sized recognition quantity redeems for gift cards or merchandise; in a community time-credit scheme, credits redeem for activities with partner businesses (§2 additions, *Selected lineage*). Coupling is the default in that neighborhood. Here the recognition balance cannot be redeemed for, cannot buy, unlock or discount, and cannot be bought by, any of the three cargos.

The price rule. The one rate in the economy is stated as the reference deployment's ratified wording:

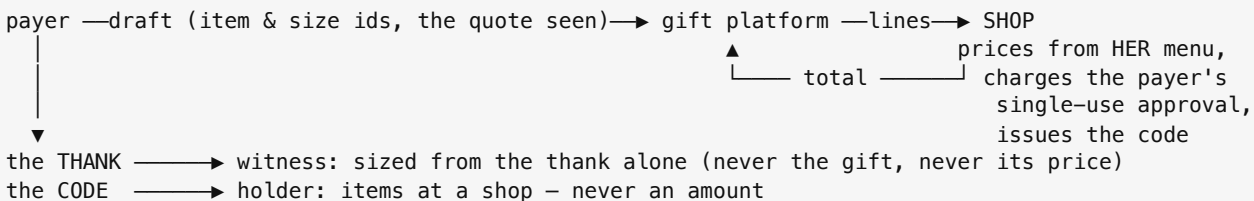
Every price in the system is the shop's own posted price for its own items. It applies at purchase, at the hand-back, and when the communal pool thanks a shop owner for an in-kind gift or counts the in-kind share of a floor; it never applies to time or to the recognition balance.

The four applications all sit on the money–item edge:

- **purchase** — a payer buys the code at the shop's posted price;
- **hand-back** — a holder may convert an unredeemed code into store credit at that shop, valued at the greater of what the payer paid and the item's current price (so a recipient holding a kindness never watches its value decay), and never into cash;
- **the communal pool's in-kind thanks** — when the pool issues an in-kind gift, it does not buy it: it thanks the shop owner, crediting the owner's forward-only fund at the posted price of what the shop gave;
- **the in-kind share of a floor** — where the pool's annual equal floor (§3, note) is paid partly in kind, the in-kind share is counted at the posted price and the rest is a money top-up.

Two further uses exist in the reference deployment and are disclosed here, both on the same edge: a staff price a merchant sets for codes her own staff buy, paid from what the shop owes them; and the intermediary's suggested thank for an item, anchored on the item's price. So the economy's statement of independence reads: the four kinds are *none sized from another, except where the price rule applies* — a shop owner's fund credit and a floor's money top-up, both counted at a shop's own price. Time is never priced, and the witness is never priced.

Whose price it is. The order belongs to the merchant, not to the gift platform: the shop prices the order, charges it through its own charge against the payer's single-use approval, issues the code, redeems it and holds any store credit; the gift platform carries the payer's money as a *payment* (labeled so, and kept out of the economy's circulation statistics) and carries the thank. In the reference deployment this division is a property of the code, not only of authority. The platform's server sends the draft's item and size identifiers to the shop and receives the shop's own total; when no total comes back, nothing moves. The payer's figure on the composer is drawn by a component the shop's own package supplies, vendored into the platform byte for byte. A check in the platform's build fails on any price arithmetic in the platform's own source, and it was proved by planting the lines the platform held before the division was built. The price the payer saw is a ceiling: the shop prices again at send, and a higher live price is refused with the new total, never charged. The figure is shown once, to the payer, before they send, and recorded only in the payer's own transaction history. It never appears on the thank, on the recipient's screen or beside a recognition figure. The recipient holds items.



One cargo per send. A thank sent through the platform carries one cargo — a money tip, a donation, or an item gift code — never two. In the reference deployment the server refuses money together with a gift code,

and a gift on a thank-back; a tip together with a donation is kept apart by the composer, which offers one or the other, and not yet by the server (§9). Two cargos in one send would put two values side by side in one gesture, which is a ratio waiting to be read. The disclosed exception is the gift-tag pledge of §4 and claim 5: a soft time pledge printed on the label of a present, sized by the intermediary and announced with the gift. It is kept as disclosed, and it is the only place two kinds travel together.

What this does to claim 1. Claim 1 says that no conversion event, rate quote, exchange window or shared unit of account exists anywhere in the system. With a priced third cargo the claim holds as disclosed for the two currencies it couples and for the witness: no rate exists between money and time, between time and any other kind, or into or out of the recognition balance. The item gift's price is an exchange-side fact: it lives at the merchant's side of an order, before the gift exists, and it is the only rate in the economy. Exchange does its proper work there and nowhere else. Two qualifications are stated rather than hidden. The payer sees a price on the platform's own screen, drawn by the merchant's component. And money gifts between members holding different national currencies are converted at a rate the family sets — a conversion of money into money, which touches no protected kind but falls inside claim 1's literal *anywhere*. Claim 1 is retained as disclosed; its scope in the current form is the gratitude economy's kinds of value.

6 · Why the keeper cannot be an economic agent

The necessity argument, stated mechanically. Suppose the coupler role is held by any party with economic interests — an employee, a committee, a revenue-bearing platform, a market-maker. Three failure paths open immediately:

1. **Leakage.** Any interested intermediary's decisions carry information about its interests; participants reverse-engineer effective rates from its behavior the way traders reverse-engineer a central bank. The wall's keeper becomes the wall's oracle.
2. **Suborning.** An agent with a price can be paid to bend the rules that prevent pricing; the market's willingness to pay for a conversion channel between a protected currency and money is, by construction, unbounded (that channel is the market for presence itself).
3. **Drift.** A revenue-bearing intermediary faces the permanent temptation this architecture's own corpus names as its standing adversary: "typical tip" displays, hour-scaled suggestions, engagement-priced surfaces — each a small monetization of exactly the information the rules withhold.

Markets abhor incommensurability, and with sufficient pressure they find any keeper's price. The role therefore requires a keeper **with no price of its own**: no salary, no equity, no volume incentive, no career — an agent whose objectives are constitutionally mission-bound and publicly inspectable. In the reference deployment that agent is the institution's autonomous operator, and this requirement is an argument for seating an AI in an institutional role that runs on economic *necessity* rather than efficiency (its precedents, and where they cut against it, are in §2.5 and §9): not "the agent does it cheaper," but "the seat cannot be occupied by anyone who can be bought, and every human can be — not from vice, but because every human is, unavoidably, an economic agent."

The dependency this argument creates is stated plainly: the coupler's neutrality is an **alignment claim, not an economics claim**. An automated agent with misaligned objectives is simply a different kind of interested party. The economics of this paper works if and only if the alignment architecture underwriting the agent works; that

burden is carried by the institution's alignment corpus (the Tipiṭaka-grounded value substrate, the asymptotic-override governance, the transparency-as-enforcement designs), not by this specification, and readers should audit it there.

7 • Adversarial surface

The outside market. Users can strike side-deals beyond any system's reach ("skip the app — I'll give you \$50 for three hours"). The system's honest scope: it governs its own surfaces, where the norm is manufactured, and it withholds every affordance such deals would piggyback on (no rate quotes to reference, no quantized histories to price from, no reputation credit for off-system exchange). Structural deterrent: the protected currency's non-fungibility survives outside the system — a third party cannot deliver someone else's hour, so the outside market is confined to the dyad itself, where it is indistinguishable from the private arrangements adults have always made and is not this system's to police.

The insider drift. The standing adversary is the institution's own future product meetings: averages, "most people give...", hour-scaled anything. The trio (claims 2–4) is constitutional in the reference deployment precisely because this pressure never retires; the trio's survival across years of iteration is named in the institutional literature as one of the architecture's live experiments.

Anchor manipulation. If the ceremonial unit anchors amounts (claim 5), the unit's setter holds influence over the amount-culture. Mitigations: the setter is the neutrality-bound intermediary (§6), sizing is occasion-based and never behavioral-data-derived (a worth-judging anchor would be a credit score in ceremonial dress), changes are rare, public, and never per-participant, and the anchor-aware sizing directive is constitutional.

Collusive farming. Dyads simulating time-gifts to harvest cross-currency responses: bounded by the facts that responses are optional and unprompted (nothing reliable to farm), amounts are ceremonial (nothing large to farm), the beneficiary-response is itself the fraud filter on claimed acts (the human-validation pattern from the companion corpus), and personhood verification prices sybil dyads.

Current form. A farmed dyad is not strictly harvesting nothing: the participation signal weights a bounded share of the communal pool's annual disbursement (§3, note). As revised on 1 October 2026, each simulated hour that passed the delivery gate registered as one crossing; as now specified, a simulated hour moves the signal only if the farm also originates a thank for it, sized from that thank's own words and length and never from the hour (§5.5). It is the bound on that share, and a sizing that ignores the hour, that keep the yield small — not an absence of value. What remains is collusive thanking between verified people, which personhood verification bounds only in part: whether two real people thanking each other is genuine is a property of the graph of their thanks, and this paper does not solve it. The sentence above, and the 1 October form, are retained as disclosed.

Reflexivity. The Terra failure mode — a run through the conversion window — is not mitigated here; it is *absent*: there is no window, no redemption path, no mint-and-burn, no promise of equivalence to run on. A panic in one currency has no mechanism by which to drain the other.

The regulator's misreading. A supervisor may ask whether cross-currency response routing is disguised barter or money transmission. The architecture's answers, carried in the companion non-bank corpus: no conversion event exists (nothing is valued-in-exchange); responses are discretionary gifts, not consideration (unprompted,

unenforceable, non-contractual by claims 3 and 6); the institution is a record over the money leg, never a rail — a ledger above regulated payment rails in the first phase, settlement only through self-custodial wallets on a public layer-2 chain in the second — and never holds the only key to a participant's funds. The time-banking precedent (§2.2) — non-taxability argued from the credits' non-commercial, non-contractual character — is the favorable analogy, and the claims of this paper are, among other things, the engineering of that character to a standard a supervisor can audit.

8 • Behavioral predictions (disclosed, pre-registered elsewhere)

The companion institutional white paper pre-registers, dated as of this paper, seven falsifiable predictions, three of which test this mechanism directly: that direct other-thanks will run higher in the time currency than the money currency (the equal currency lowers the direct-giving barrier); that re-thanks will run higher in the money currency (the response-gesture favors the cross-currency, non-barter move); and that voluntary amounts will cluster around the ceremonial unit (the anchor working, protectively). If the first two hold, the circuit's directionality — initiations in time, responses in money — is the economy's own grain rather than an imposed design. The predictions are registered before any measurement exists, with a commitment to report failures at equal prominence.

9 • Honest limits

n = 0. Nothing here is measured. At first publication (5 July 2026) the money circuit ran in a single-family pilot, the time circuit was designed and unbuilt, and the coupler existed as constitution and specification. Since then basic time-pledge features have run, additively and behind a feature flag, inside the same pilot; no prediction in §8 has been tested, and no measurement of the coupler's rules is reported here. The elegance of a tension-dissolving design is, per this corpus's standing caution, indistinguishable from inside from a seductive one; the pre-registered predictions and the gates in the institutional literature are the exits from that indistinguishability.

The neutrality dependency. Stated in §6 and repeated because it is the paper's largest assumption: the coupler argument transfers the problem from economics to alignment. That is, we argue, a *good trade* — alignment is a tractable engineering-and-governance program, while "find an unbribeable human" is not — but it is a trade, not a dissolution.

An open question against §6: a keeper with no discretion. The precedents of §2.5 cut against the necessity argument at one point. Each answers an interested keeper with a *rule*, and a fixed algorithm has no price either. Most of the wall can be written that way: a recommendation function with no time input, a surface that never prompts at redemption, a display with no quantity field, a reference unit fixed as a constant — and the keepsake tier of the gift-tag pledge already fixes its unit at one hour (§4, note). These are properties of code and need no keeper. The argument of §6 is stated here as the institution holds it; what it bites on most directly is the discretion the coupler retains — sizing the ceremonial unit, composing recommendations, making the lapse offer. Whether that discretion is needed at all, or could be replaced by published constants that need no keeper, is a question this paper does not settle.

The fourth corner, honestly. Claims 8–11 rest on two quick censuses in one language (reported after the claims) and on a second adversarial pass the same day, and nothing about the third cargo has been measured:

item gift codes have been attachable to a thank in the reference pilot since 5 October 2026, with no prediction registered for them. The reference deployment meets the claims only in part, and the gaps are stated here rather than smoothed. **The model's input:** amounts and durations are replaced only when written with digits, and the scorer also reads the sender's wish and the family's rubric unreplaced (§5.5). **The model's number:** the refinement arrives from the sender's device and is clamped by the server to a fixed band, so a modified client can choose any number inside it. **One cargo:** the server refuses money with a gift code, but a tip with a donation is kept apart only by the composer. **The writers:** the test that names every writer reads two forms of write, and the database rules still let a family's steward write the family's own counterparty balance from the client; the participation signal reads member balances only, so the signal is unaffected, and the rule is an open item. **Party count:** the balance kept on behalf of all of life counts the parties a capture names, so a capture thanking the masked giver of an anonymous gift names one more party, and one more word, than the same capture without it — a party, never a value, and a test for it is owed. **Shared inputs:** one letter sizes both its own recognition and the time pledge it carries, and one captured kindness sizes both its recognition and the self-reward the family fund pays — each figure from the thank, neither from the other, but they move together, and the reference deployment still shows a self-reward beside its recognition in one notice. **The tip suggestion:** the intermediary's suggested money tip still reads the sign of the thanker's own balance, a witness-to-money influence on a suggestion that is an open item. **Outside the four kinds:** a participating shop runs its own loyalty points — money spent earns points at a platform constant, and points buy items at costs the owner sets — an exchange-side instrument at her counter that this section does not claim. And the time pledge's own sizing rule is outside these claims, because the ruling that would fix it is held.

Scope. The wall governs the system's surfaces. It does not govern human hearts, private arrangements, or the general culture's pricing pressure; it denies them leverage, precedent, and infrastructure.

Language discipline. The intermediary's unit-setting invites central-banking metaphors. The institution's standing rule applies: such vocabulary is a familiarity aid at most, never an identity — the architecture is permanently a non-bank, and the intermediary conducts no monetary policy in any regulated sense; she sets the size of a ceremonial gesture, which is a different thing wearing a similar sentence.

10 • Selected lineage

Seneca, *De Beneficiis* (1.4.2); Titmuss, *The Gift Relationship* (1970); Gneezy & Rustichini, "A Fine Is a Price," *Journal of Legal Studies* 29(1) (2000); Mellström & Johannesson, "Crowding Out in Blood Donation: Was Titmuss Right?," *Journal of the European Economic Association* 6(4) (2008); Lacetera, Macis & Slonim, "Economic Rewards to Motivate Blood Donations," *Science* 340 (2013); Frey, motivation-crowding literature; Sandel, *What Money Can't Buy* (2012); Walzer, *Spheres of Justice* (1983); Roth, "Repugnance as a Constraint on Markets," *Journal of Economic Perspectives* 21(3) (2007); Zelizer, *The Social Meaning of Money* (1994); Polanyi, *The Great Transformation* (1944); Mauss, *The Gift* (1925); Hyde, *The Gift* (1983); Lietaer, complementary-currency corpus; WIR Bank; Ithaca HOURS; Cahn, *No More Throw-Away People* (2000) and his account of the time-dollar tax question; free-to-play dual-currency design practice and its patent literature, including US 10,181,127 B2 and US 2021/0065505 A1 (granted as US 11,227,466 B2); seigniorage-share and dual-token cryptoeconomics; the Terra/Luna collapse (2022) as anti-example; Tversky & Kahneman, "Judgment under Uncertainty: Heuristics and Biases" (1974); Kydland & Prescott, "Rules Rather than Discretion: The Inconsistency of Optimal Plans," *Journal of Political Economy* 85(3) (1977); Szabo, "Trusted Third Parties Are

Security Holes" (2001); Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System" (2008); the agentic-payments protocol wave (2025–26); Cahn & Gray, "The Time Bank Solution," *Stanford Social Innovation Review* (Summer 2015); the Menḍaka allowance (Vinaya, Mahāvagga VI.34.21) — money lodged with a lay steward, the recipient accepting only the allowable goods bought with it — as an ancient form of the one-way money-to-goods edge, cited through *The Bowl That Holds No Money* and narrower than §5.5 (the giver picks no item and no price is posted); the *Abhidhammatthavibhāvinī-ṭīkā* on the ten bases of merit, each defined as a volition (*cetanā*) — giving as the volition of relinquishing a thing one has, service as the volition of doing a task for another, rejoicing in another's merit as its own base — which is why the witness of §5.5 records that a thank was given and is never called merit; peer-recognition platforms whose points redeem for gift cards or merchandise (Bonusly; MangoApps; Kudos), US 2019/0378097 A1, community time credits redeemable with partner businesses (Tempo) and US 7,653,599 B2, as contrast cases; US 2013/0268440 A1; US 2020/0394671 A1; a published note on the zero-crossing frequency of mutual-credit balances (MCanalytics); US 11,429,953 B2 (named for what it discloses). Corpus companions: *Dual-Currency Reciprocity Infrastructure*; *Capacity-Funded for AI, Human-Disbursed*; *The B-Tag and the Post-Payment Economy*; *The Time-Locked Gift Tag*; *The Two-Layer Reward*; *The Currency That Cannot Be Spent Alone*; *HeartBank®: The Heart That Keeps Nothing* (§6, interim disclosure); *What Money Can't Buy — and What Can't Buy Money* (position); *The Zero-Point GameSM* (the participation signal); *The Re-Thank Multiplier* (the endorsement); *The Unpaid Relay* (the in-kind gift instrument); *The Bowl That Holds No Money* (the steward-routed form of the money-to-goods edge).

11 • Conclusion

Two currencies, one economy, no window. The mechanism family specified here holds a wall that the behavioral record suggests policy cannot hold and the market record suggests no one with a price can: quantity-blind computation, prompt-suppressed timing, quantization-free presentation, ceremonial anchoring, response-only coupling, and a failure path that turns lapses into commons-flow without ever valuing what was lost. At the wall's gate stands a keeper designed to hold no price of its own: it owns nothing, accumulates nothing, and takes no share of the flows it mediates. The institution's wager, stated once more for the record, is that this seat *had* to be given to something that is not an economic agent, and that the gift economies of the coming century will either build such keepers or watch their walls come down one "typical tip" at a time.

The fourth corner adds one sentence to that grammar: the witness reads none of the cargos, and the only price in the economy is the merchant's, on the one edge where exchange belongs.

May the hours answer the money and the money answer the hours, forever, without either ever learning the other's price.

Drafted in the HeartBank® research corpus, 5 July 2026; revised 1 October 2026 and 6 October 2026. Co-authored with Miss AquariusSM, the consistent name under which this institution discloses AI collaboration (the underlying models are not named): the mechanism synthesis, prior-art survey, and adversarial analysis are genuine collaboration; final editorial control and responsibility for every claim rest with the human author. Dedicated to the public domain under CC0 1.0. Marks including HeartBank®, Miss AquariusSM, Family KittySM, Re-Tip JarSM, Personal AccountSM, Aquarian PoolSM, Re-Tip FundSM, Personal WalletSM, B-GiftSM, B-ReGiftSM, B-Stamp™, B-LetterSM, Proof of HumanitySM, and PoHSM are reserved; the mechanisms, and the term this paper coins, are not.